CFA LEVEL I | QUANTITATIVE METHODS

Quant Notes

Decision Trees, Formulas & Rule Books

A consolidated, exam-ready reference covering compounding conversions, return measures (TWRR, MWRR, arithmetic/geometric/harmonic mean), interest rate components, hypothesis testing, regression, probability, sampling, research biases, confidence intervals, and required/expected/realized return — organized as quick-recall decision trees, formulas, and a fully worked exam question.

Compiled Study Notes · Not an official CFA Institute publication

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1 Annualized Compound Return — Periodic vs Continuous Compounding

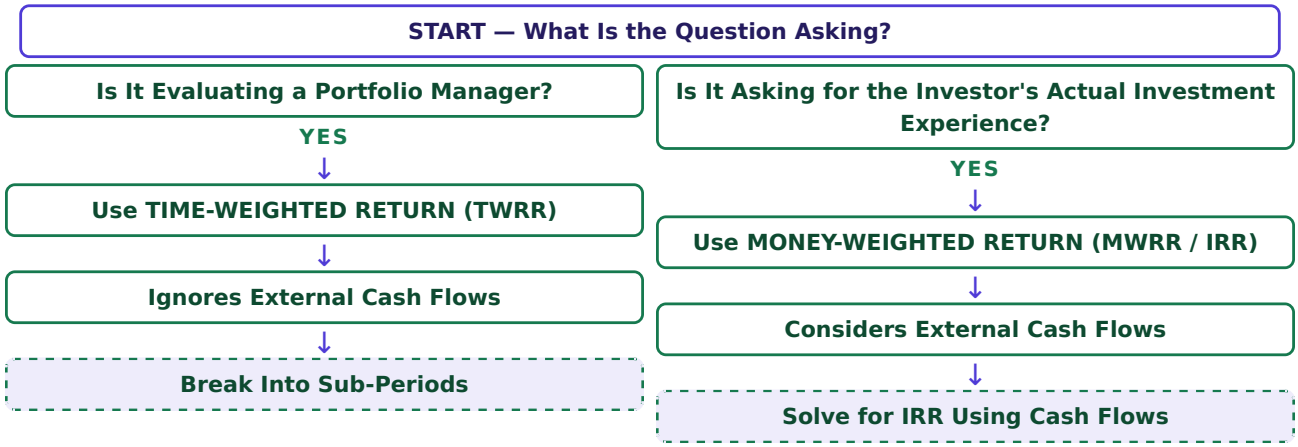
> **Annualized Compound Return (Effective Annual Return)**
> Return converted to a one-year equivalent using compounding.

Converting Between Periodic and Continuous Compounding

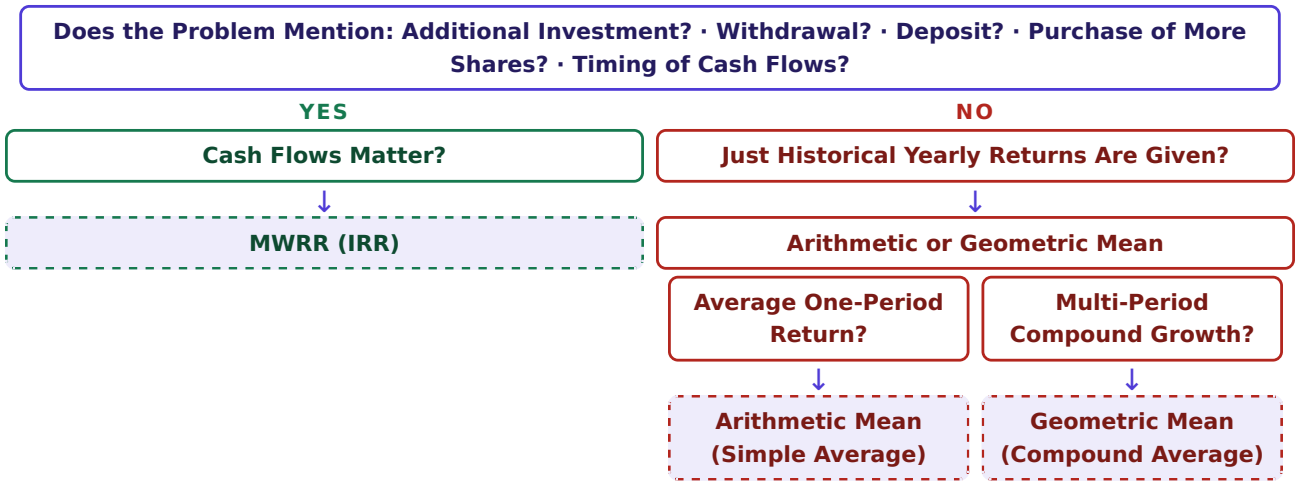
where EAR is the Effective Annual Rate.

Need to Find	Formula
Future Value	$FV = PV \cdot e^{rt}$
Present Value	$PV = FV \cdot e^{-rt}$
Continuous Rate	$r = \ln(FV / PV) / t$
Effective Annual Rate	$EAR = e^{r} - 1$
Continuous Rate from EAR	$r = \ln(1 + EAR)$

2 CFA Level I — Return Measures Decision Tree



Continued — Does the Problem Mention Cash Flow Timing?



3 Components of a Nominal Interest Rate

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Nominal Interest Rate = Real Risk-Free Rate + Expected Inflation + Risk Premiums
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where Risk Premiums may include: Default risk premium · Liquidity premium · Maturity premium

> Think of it as building up the required return layer by layer.

Component	Compensation For
Real Risk-Free Rate	Compensation for delaying current consumption
Expected Inflation	Compensation for loss of purchasing power
Default Risk (Risk Premium)	Compensation for possibility of issuer default
Liquidity Risk (Risk Premium)	Compensation for holding an illiquid asset
Maturity (Interest Rate) Risk (Risk Premium)	Compensation for holding a longer-maturity bond

Ask Yourself

"Why should someone wait instead of consuming today?" → They earn the **Real Risk-Free Rate**.
"Why should someone hold an illiquid asset?" → **Liquidity Premium**.
"Why hold a 30-year bond instead of a 1-year bond?" → **Maturity Premium**.

4 Worked Exam Question — Hypothesis Test on Difference in Means

Question

An analyst performs a hypothesis test concerning the difference between the mean returns of two portfolios, assuming normally distributed populations with unknown but equal variances. If the analyst decides to change the hypothesized difference in mean returns from 0% to 1%, which of the following will change?

This is a very common CFA Level I hypothesis testing concept. The exam is testing whether you know which parts of the t-test formula depend on the hypothesized difference.

Step 1 — Identify the Test

The question states: Two population means · Normal populations · Unknown but equal variances.

This tells us to use the **pooled two-sample t-test**.

$$t = [(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)_0] / \sqrt{(s_p^2 (1/n_1 + 1/n_2))}$$

where: $(\mu_1 - \mu_2)_0$ = hypothesized difference | s_p = pooled standard deviation | n_1, n_2 = sample sizes

Step 2 — What Changed?

The hypothesized difference changes from 0% to 1%. That means only this part changes: $(\mu_1 - \mu_2)_0$. Everything else stays exactly the same.

Step 3 — Which Part of the Formula Changes?

$$\text{Before: } (\bar{X}_1 - \bar{X}_2) - 0$$
$$\text{After: } (\bar{X}_1 - \bar{X}_2) - 1\%$$

Since the Numerator Changes

✅ The test statistic changes.

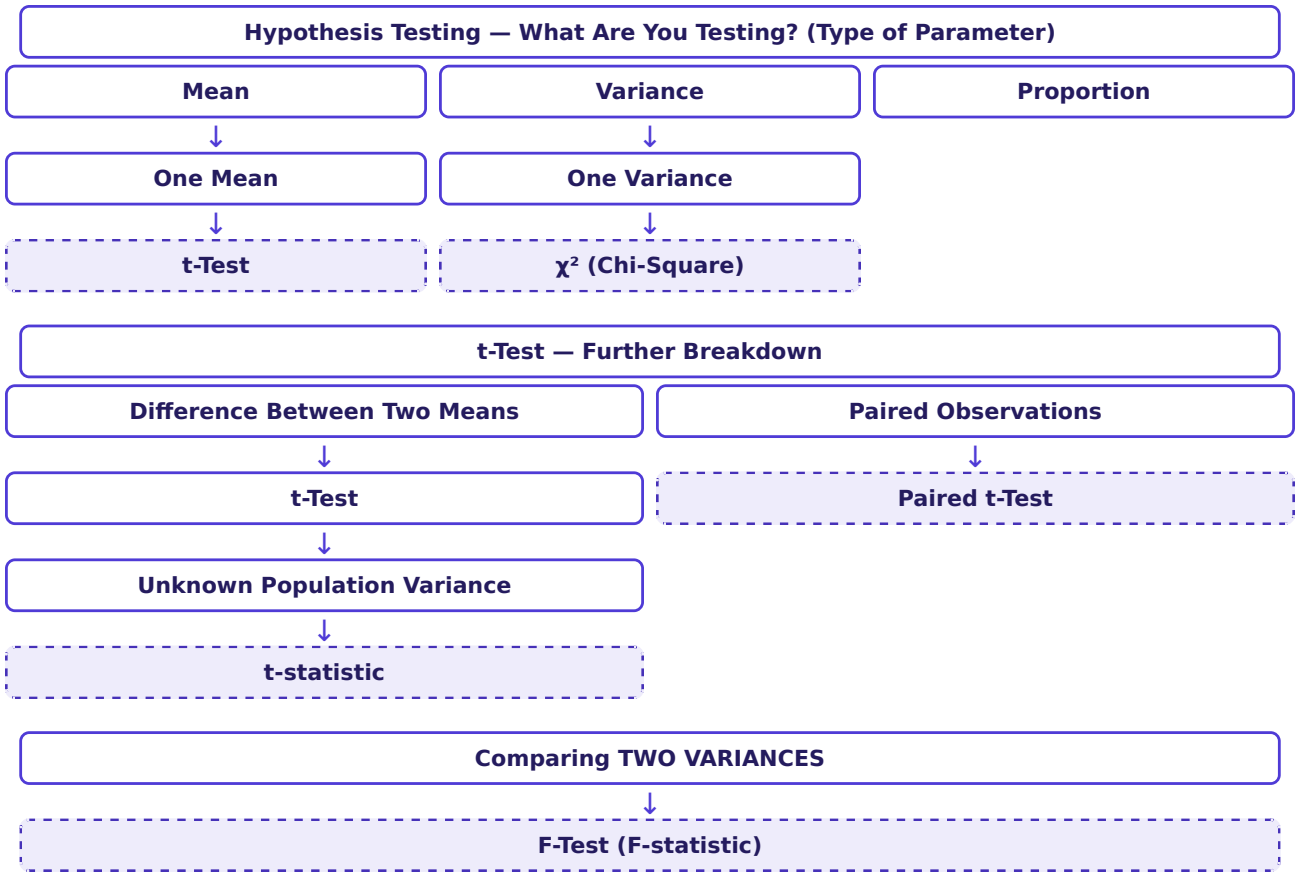
Correct Answer

A. The value of the test statistic

❌ Why the Other Options Are Wrong

Option	Why It's Wrong
B. Degrees of Freedom ❌	For a pooled t-test, df = $n_1 + n_2 - 2$. This depends only on the sample sizes. Changing the hypothesized mean difference has no effect.
C. Pooled Estimate of the Common Population Variance ❌	The pooled variance is computed from the sample variances. It depends only on sample variances and sample sizes — not on the hypothesized mean difference.

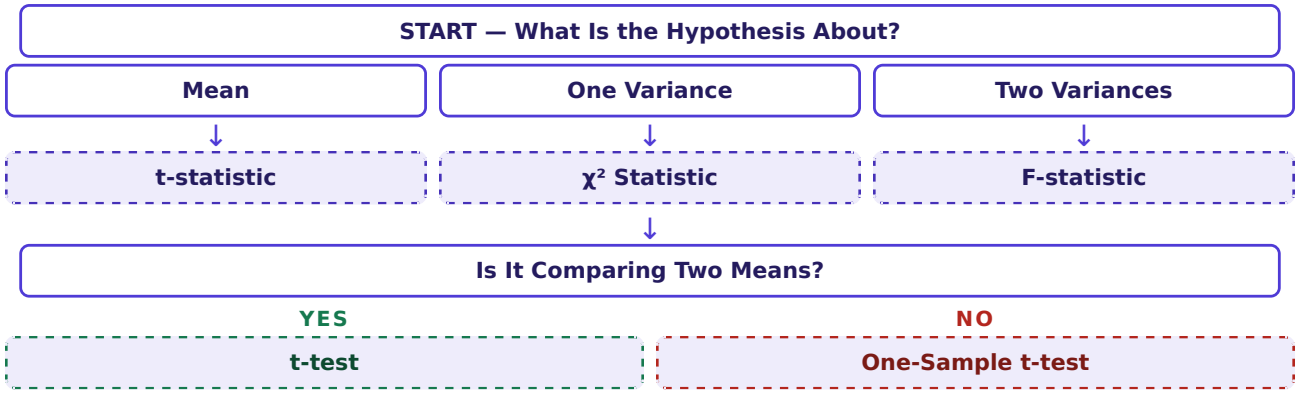
5 CFA Knowledge Graph — Choosing the Correct Test Statistic



Summary of Test Statistics

Parameter	Test Statistic
Mean	t
One Variance	χ^2
Two Variances	F

🎯 Decision Tree (CFA Exam Version)



Parameter	Test Statistic	CFA Trigger Words
Mean	✅ t	Mean, Average, Expected Return
Difference Between Means	✅ t	Compare Means
Paired Mean Difference	✅ t	Before/After, Matched Pairs
One Variance	✅ χ^2	Variance, Volatility
Two Variances	✅ F	Compare Variability

6 When to Use Nonparametric Procedures

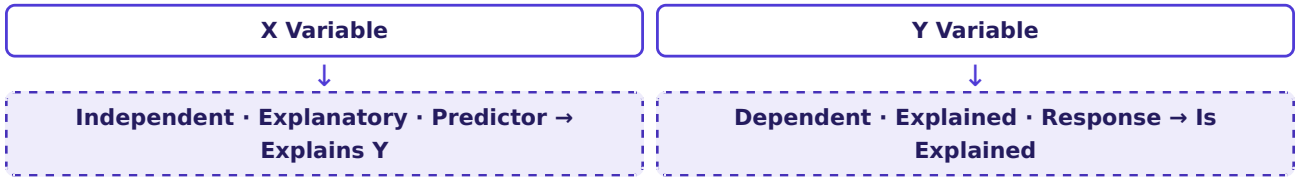
> We perform nonparametric procedures in four situations:

1. When the data we use do not meet distributional assumptions.
2. When there are outliers.
3. When the data are given in ranks or use an ordinal scale.
4. When the hypotheses we are addressing do not concern a parameter.

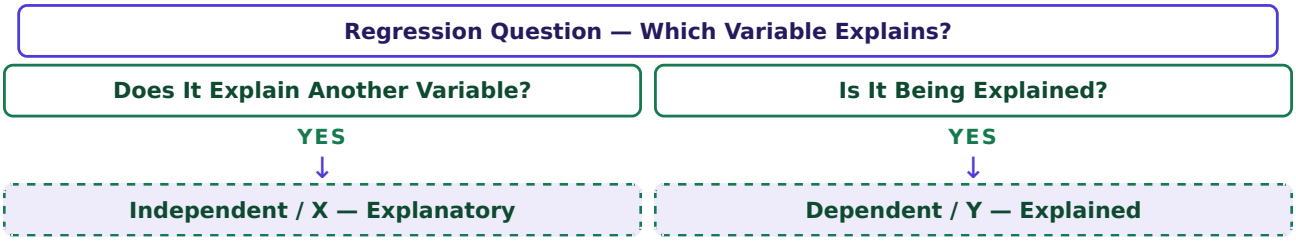
> **Exam Note**
> A dataset containing outliers is Situation 2 — one in which a nonparametric test would be appropriate.

7 CFA Rule Book — Regression Variables

Simple Linear Regression: $Y = b_0 + b_1X + \varepsilon$



🎯 Decision Tree



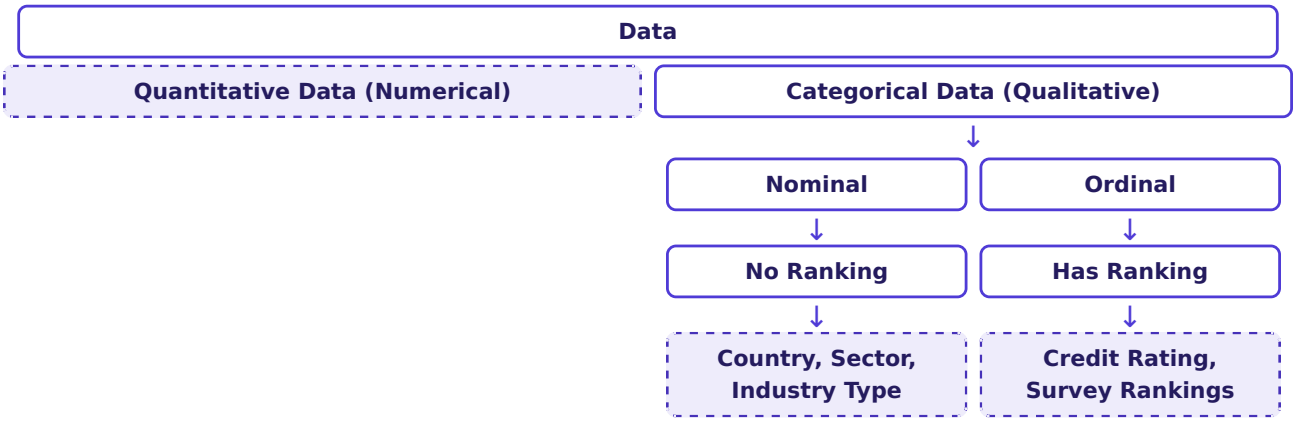
8 Standard Error of the Estimate (SEE)

Measures

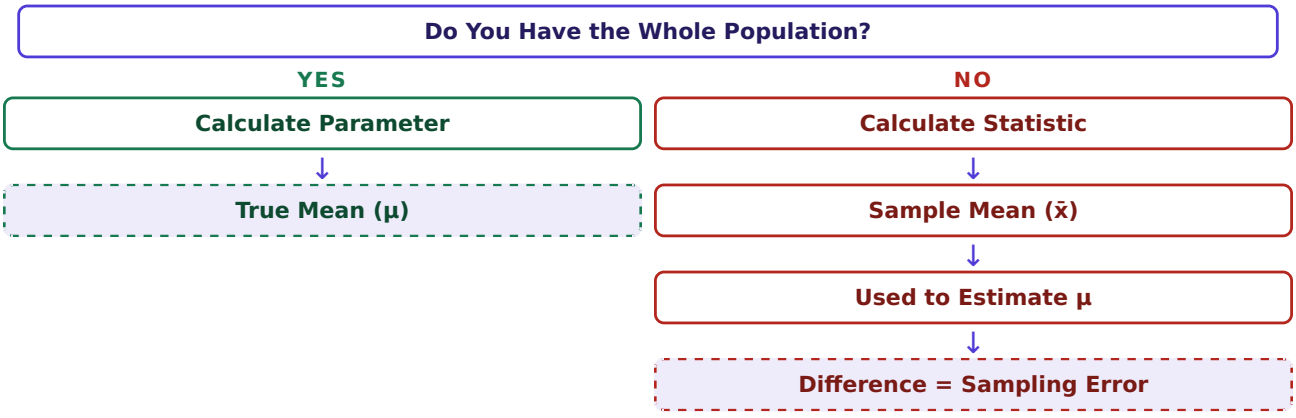
The typical size of the regression residual (prediction error).

$$SEE = \sqrt{\Sigma(Y - \hat{Y})^2 / (n - 2)} = \sqrt{SSE / (n - 2)}$$

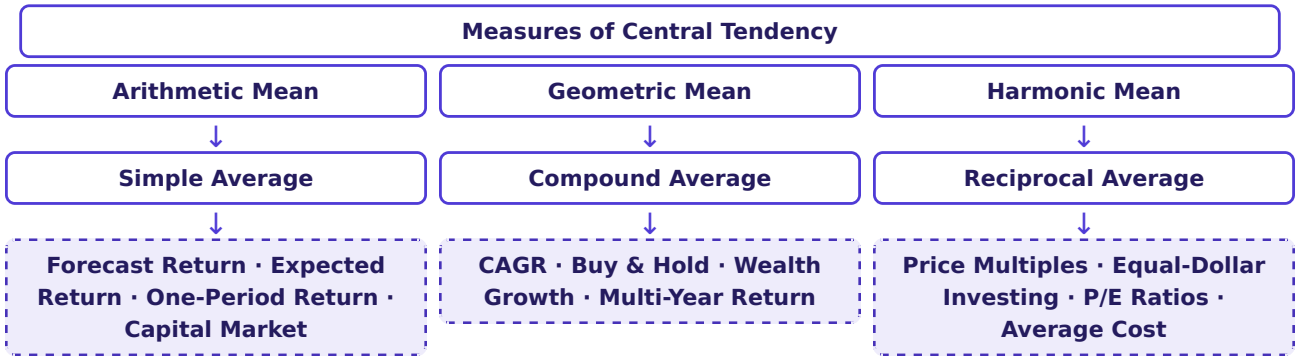
9 Understanding Categorical Data



10 Sampling and Estimation



11 One-Page Rule Book — Measures of Central Tendency



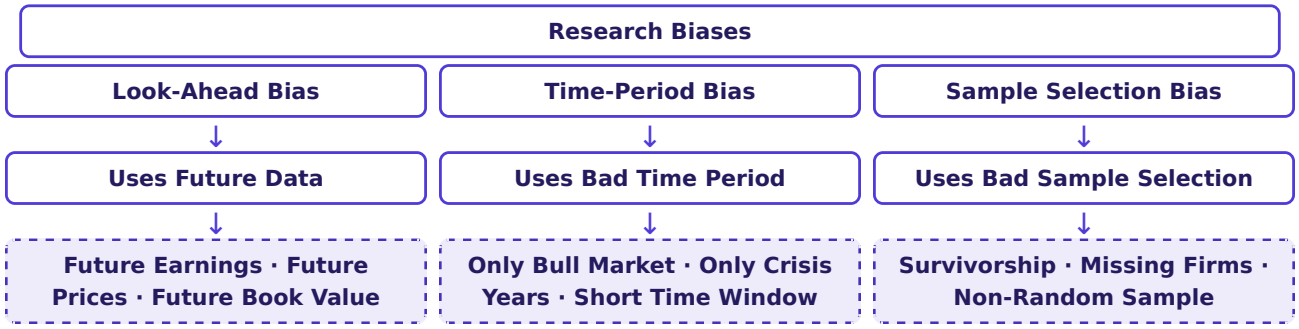
Read the Question — Sequential Checklist

Step	Question to Ask	If Yes →
1	Does it ask about FUTURE return?	**Arithmetic Mean**
2	(If no) Does it ask about compound growth?	**Geometric Mean**
3	(If no) Does it involve ratios/multiples?	**Harmonic Mean**
4	(If no to all above)	**Look for Median/Mode**

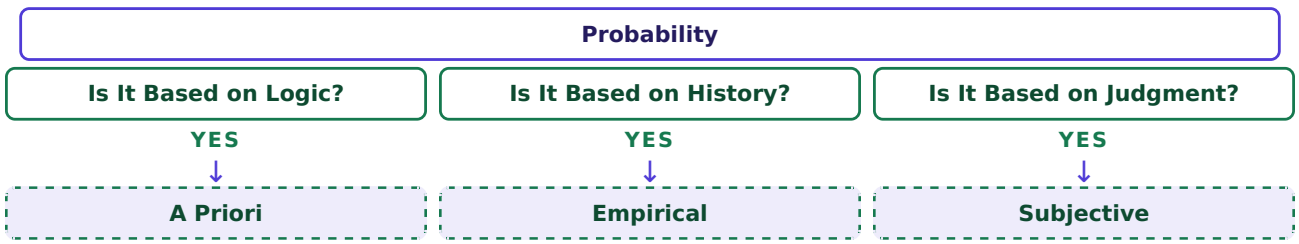
Work Top to Bottom

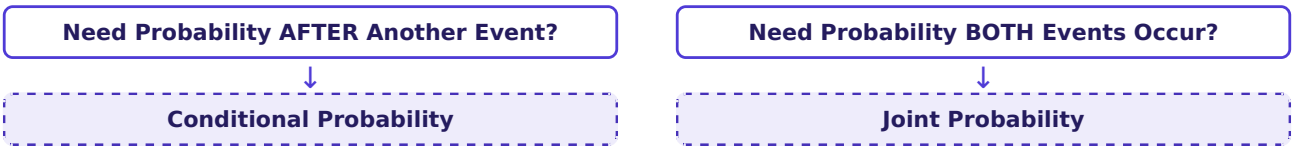
Check each question in order — the first "Yes" gives you the answer. Only if every answer is "No" do you fall through to Median/Mode.

12 CFA Rule Book — Common Research Biases

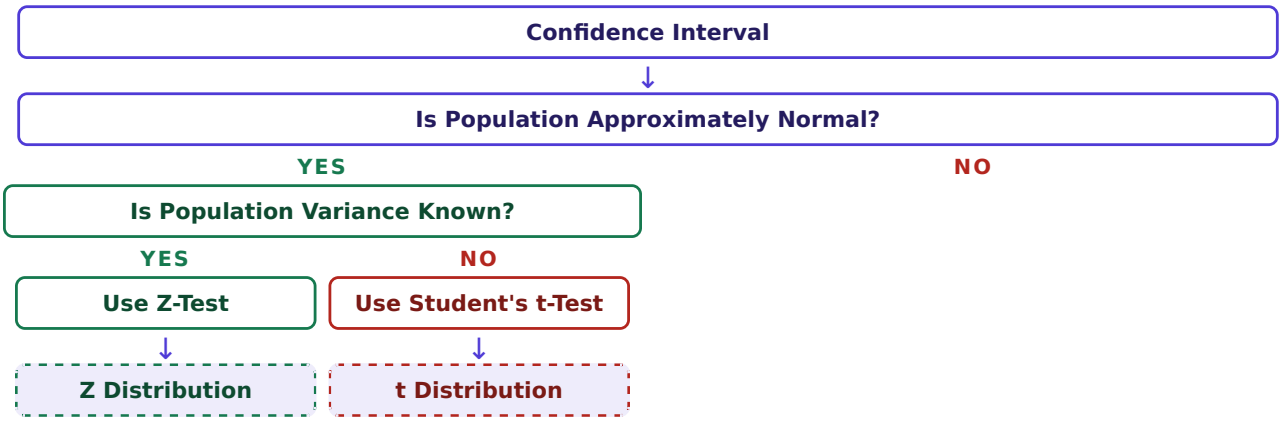


13 Types of Probability

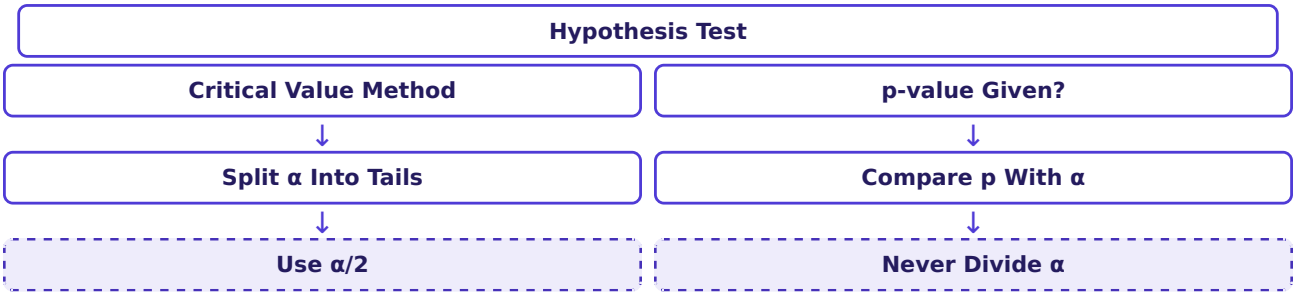




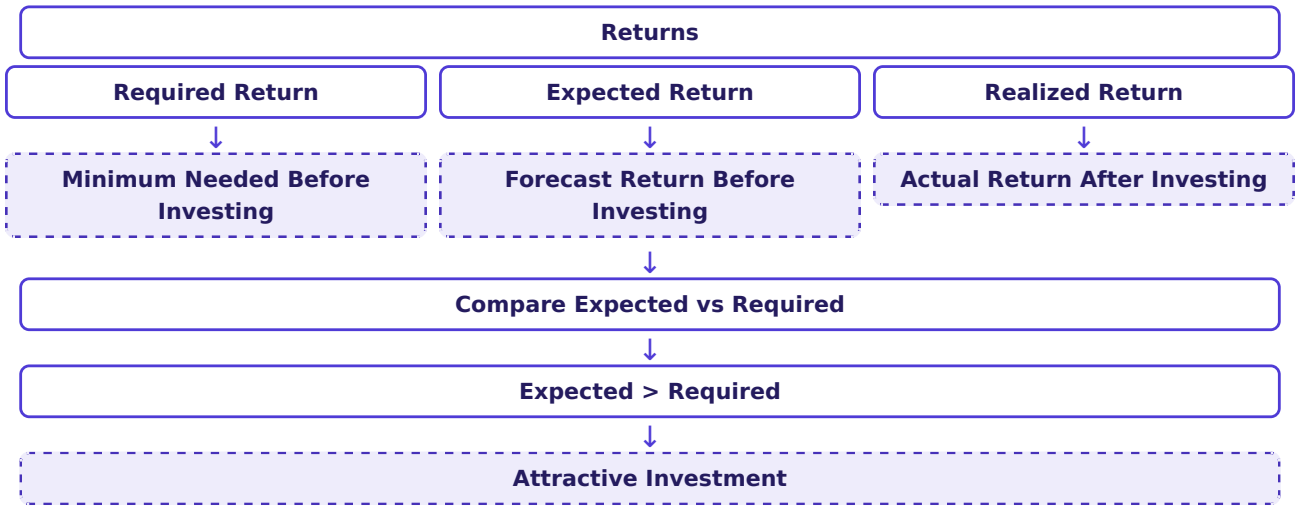
14 Confidence Interval for the Population Mean



15 Two Methods in Hypothesis Testing



16 One-Page Rule Book — Required, Expected & Realized Return



Cash Flow Based?

